

From controllability and observability to the regularity of Theorem 1: a self-contained proof for the LQG co-design SDPs

ContEst supplementary note

July 2, 2026

Abstract

Theorem 1 of the ContEst paper computes the exact design gradient of a convex inner problem by contracting the model Jacobians against the dual variables the inner solver already returns. It holds under Assumption 1, a set of conic-program regularity conditions — convexity and C^1 data with a Slater point (A1), uniqueness of the primal minimizer (A2), and nondegeneracy / strict complementarity so that the dual is unique (A3) — which in the paper are simply *assumed* and checked *a posteriori* by finite differences. This note removes that gap in the linear-quadratic-Gaussian (LQG) case. We prove that for both the control SDP (8) and the estimation SDP (9), Assumption 1(A1)–(A2) is a *consequence* of the classical hypotheses that (A_θ, B_θ) is stabilizable, (A_θ, C_θ) is detectable, $W, V \succ 0$, $R \succ 0$, and $Q \succeq 0$ with $(A_\theta, Q^{1/2})$ detectable (the last automatic when $Q \succ 0$). We construct the Slater points explicitly, identify the unique primal optimizer with the stabilizing solution of the corresponding discrete algebraic Riccati equation (DARE), and show the optimal value is a C^1 function of θ . From this we derive the operational conclusion of Corollary 1 in a form that does *not* even require (A3): the envelope gradient read from *any* optimal dual returned by the solver equals the true $\nabla_\theta V$. Strict complementarity (A3) is then shown to be the exact condition under which the dual is moreover *unique*, recovering Assumption 1 verbatim. The result turns Corollary 1 from “differentiable under regularity we hope holds” into “differentiable under standard system-theoretic hypotheses,” making the paper self-contained and citable.

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1 Setup and statement

We work in discrete time. Fix a design point θ in an open set $\Theta_0 \subseteq \mathbb{R}^{r_\theta}$ and consider the linear model of Section 5 of the paper,

$$x_{k+1} = A_\theta x_k + B_\theta u_k + w_k, \quad y_k = C_\theta x_k + v_k, \quad (1)$$

with $x_k \in \mathbb{R}^n$ ($n = r_x$), $u_k \in \mathbb{R}^m$, $y_k \in \mathbb{R}^p$, and zero-mean white w_k, v_k of covariances $W = W(\theta)$ and $V = V(\theta)$. The stage weights are $Q = Q(\theta) \succeq 0$ and $R = R(\theta) \succ 0$. We use $\mathbb{S}^q$ for the symmetric $q \times q$ matrices, $X \succ 0$ ($X \succeq 0$) for positive (semi)definiteness, and $\langle X, Y \rangle = \text{tr}(X^\top Y)$.

The paper reduces the LQG inner value, by the separation principle applied *within the inner solve*, to two semidefinite programs. We restate them.

Control SDP (Eq. (8)).

$$\begin{aligned} J_{\text{cont}}(\theta) = \min_{\Sigma, L, Z_0} \quad & \text{tr}(Q\Sigma) + \text{tr}(RZ_0) \\ \text{s.t.} \quad & \Sigma \succ 0, \quad \begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix} \succeq 0, \\ & \underbrace{\begin{bmatrix} \Sigma - W & A_\theta \Sigma + B_\theta L \\ (A_\theta \Sigma + B_\theta L)^\top & \Sigma \end{bmatrix}}_{=: \mathcal{G}_c(\Sigma, L; \theta)} \succeq 0, \end{aligned} \quad (2)$$

with optimal gain $K^* = L^*(\Sigma^*)^{-1}$.

Estimation SDP (Eq. (9)).

$$\begin{aligned} J_{\text{est}}(\theta) = \min_{P^\varepsilon, F, Z_1} \quad & \text{tr}(WP^\varepsilon) + \text{tr}(VZ_1) \\ \text{s.t.} \quad & P^\varepsilon \succ 0, \quad \begin{bmatrix} Z_1 & F^\top \\ F & P^\varepsilon \end{bmatrix} \succeq 0, \\ & \underbrace{\begin{bmatrix} P^\varepsilon - Q & A_\theta^\top P^\varepsilon - C_\theta^\top F^\top \\ P^\varepsilon A_\theta - FC_\theta & P^\varepsilon \end{bmatrix}}_{=: \mathcal{G}_e(P^\varepsilon, F; \theta)} \succeq 0. \end{aligned} \quad (3)$$

Both are instances of the generic conic program of the paper,

$$V(\theta) = \min_z \phi(z, \theta) \quad \text{s.t.} \quad g(z, \theta) = 0, \quad G(z, \theta) \in \mathcal{K}, \quad (4)$$

with $\mathcal{K}$ the semidefinite cone, Lagrangian $L(z, \mu, S, \theta) = \phi + \langle \mu, g \rangle + \langle S, G \rangle$, and optimal triple (z^*, μ^*, S^*) . Theorem 1 states that under Assumption 1,

$$\nabla_\theta V(\theta) = \nabla_\theta L(z^*, \mu^*, S^*, \theta) = \partial_\theta \phi + \langle \mu^*, \partial_\theta g \rangle + \langle S^*, \partial_\theta G \rangle, \quad (5)$$

the partials taken with (z^*, μ^*, S^*) held fixed. Assumption 1 requires, on Θ_0 : (A1) convexity, jointly C^1 data, and Slater; (A2) a unique primal minimizer; (A3) nondegeneracy and strict complementarity, hence a unique dual; (A4) θ enters only through finitely many C^1 data objects $D(\theta)$.

We prove the following.

Assumption 1 (System-theoretic hypotheses). On the open neighborhood Θ_0 of θ :

- (S1) the maps $\theta \mapsto (A_\theta, B_\theta, C_\theta, W, V, Q, R)$ are C^1 ;
- (S2) (A_θ, B_θ) is stabilizable and (A_θ, C_θ) is detectable;
- (S3) $W(\theta) \succ 0$, $V(\theta) \succ 0$, $R(\theta) \succ 0$, $Q(\theta) \succeq 0$, and $(A_\theta, Q^{1/2})$ is detectable (automatic when $Q \succ 0$).

Theorem 1 (Main). *Under Assumption 1, for each of the SDPs (2) and (3) and every $\theta \in \Theta_0$:*

- (i) (A1) and (A4) hold. *The program is convex with jointly C^1 data, θ enters only through $D(\theta) = (A_\theta, B_\theta, C_\theta, W, V, Q, R)$, a Slater point exists, and consequently strong duality holds with an attained primal–dual solution.*
- (ii) (A2) holds. *The primal minimizer is unique and equals the stabilizing solution of the associated DARE; in particular $\Sigma^* \succ 0$ (resp. $P^{\varepsilon*} \succ 0$).*
- (iii) Value differentiability and Corollary 1. *$V(\cdot) \in C^1(\Theta_0)$, and for every optimal dual (μ^*, S^*) the right-hand side of (5) equals $\nabla_\theta V(\theta)$. Hence the LQG design gradients of Corollary 1 — Eqs. (12)–(13) and (16)–(17) of the paper — computed from the duals the solver returns, are exact.*
- (iv) (A3) and Assumption 1 verbatim. *If, in addition, the primal–dual solution is strictly complementary and primal nondegenerate, then the dual (μ^*, S^*) is unique, so Assumption 1(A1)–(A4) holds in full and Theorem 1 applies as stated.*

Part (iii) is the operationally decisive statement: it validates Corollary 1 under (S1)–(S3) *alone*, i.e. under controllability/observability-type hypotheses, without invoking strict complementarity. Part (iv) records that (A3) is exactly the extra ingredient that upgrades “the returned dual gives the right gradient” to “the dual is unique,” which is Assumption 1 as written.

The remainder proves Theorem 1. Section 2 disposes of (A1)/(A4) except Slater; Section 3 constructs Slater points from (S2)–(S3); Section 4 identifies the primal optimum with the DARE solution and proves uniqueness; Section 5 proves value differentiability and the dual-invariant gradient formula; Section 6 treats strict complementarity and dual uniqueness. Because (3) is the transpose dual of (2), we prove everything for the control SDP and transfer to the estimation SDP by an explicit substitution (Section 7).

2 Convexity, smoothness of the data, and (A4)

Lemma 1. *For each fixed θ , (2) is a convex program: the objective is linear in (Σ, L, Z_0) and the three constraints are, respectively, an open linear-matrix strict inequality and two linear-matrix inequalities in (Σ, L, Z_0) . Its data (ϕ, g, G) are affine in the decision variables and depend on θ only through $D(\theta) = (A_\theta, B_\theta, W, Q, R)$; under (S1) they are jointly C^1 in (z, θ) . The identical statement holds for (3) with $D(\theta) = (A_\theta, C_\theta, W, V, Q)$.*

Proof. The objective $\text{tr}(Q\Sigma) + \text{tr}(RZ_0)$ is linear. The matrices $\begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix}$ and $\mathcal{G}_c(\Sigma, L; \theta)$ are affine functions of (Σ, L, Z_0) whose coefficients are C^1 in θ by (S1); requiring them to lie in the PSD cone $\mathcal{K}$ is a convex constraint. The strict constraint $\Sigma \succ 0$ defines an open convex set. All θ -dependence is carried by the coefficient matrices $A_\theta, B_\theta, W, Q, R$, giving (A4). C^1 jointness in (z, θ) is immediate because the entries are polynomials in z with C^1 coefficients in θ . $\square$

Thus (A1) reduces to Slater’s condition, and (A4) is already established. We now produce a Slater point.

3 Slater's condition from stabilizability and detectability

The one place where the system-theoretic hypotheses do genuine work is strict feasibility. We give an explicit strictly feasible triple, so Slater's condition is not merely asserted but exhibited.

Lemma 2 (Slater for the control SDP). *Assume (S2)–(S3). Then (2) has a Slater point: there exist (Σ, L, Z_0) with $\Sigma \succ 0$, $\begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix} \succ 0$, and $\mathcal{G}_c(\Sigma, L; \theta) \succ 0$ (all constraints strict).*

Proof. By stabilizability of (A_θ, B_θ) pick K with $A_{\text{cl}} := A_\theta + B_\theta K$ Schur (spectral radius < 1). Fix any $\varepsilon > 0$. Because A_{cl} is Schur and $W + \varepsilon I \succ 0$, the discrete Lyapunov equation

$$\Sigma = A_{\text{cl}} \Sigma A_{\text{cl}}^\top + W + \varepsilon I \quad (6)$$

has a unique solution $\Sigma = \sum_{k \geq 0} A_{\text{cl}}^k (W + \varepsilon I) A_{\text{cl}}^{\top k} \succ 0$. Set $L := K \Sigma$ and $Z_0 := K \Sigma K^\top + \varepsilon I$. We check the three constraints.

(a) $\Sigma \succ 0$ by construction.

(b) The Schur complement of the block $\begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix}$ with respect to $\Sigma \succ 0$ is $Z_0 - L \Sigma^{-1} L^\top = K \Sigma K^\top + \varepsilon I - K \Sigma \Sigma^{-1} \Sigma K^\top = \varepsilon I \succ 0$; hence the block is $\succ 0$.

(c) In $\mathcal{G}_c$ we have $A_\theta \Sigma + B_\theta L = (A_\theta + B_\theta K) \Sigma = A_{\text{cl}} \Sigma$. Its Schur complement with respect to the $(2, 2)$ block $\Sigma \succ 0$ is

$$(\Sigma - W) - (A_{\text{cl}} \Sigma) \Sigma^{-1} (A_{\text{cl}} \Sigma)^\top = \Sigma - W - A_{\text{cl}} \Sigma A_{\text{cl}}^\top \stackrel{(6)}{=} \varepsilon I \succ 0,$$

so $\mathcal{G}_c(\Sigma, L; \theta) \succ 0$. (Note $\Sigma - W \succ 0$ as well, since $\Sigma - W = A_{\text{cl}} \Sigma A_{\text{cl}}^\top + \varepsilon I \succ 0$.) $\square$

Corollary 1 (Strong duality and attainment). *Under (S2)–(S3), (2) is bounded below, strong duality holds, and the dual optimum is attained; the primal optimum is attained as well (Section 4). Consequently Assumption 1(A1) holds.*

Proof. Boundedness below: for feasible (Σ, L, Z_0) , $\Sigma \succeq 0$ and $Q \succeq 0$ give $\text{tr}(Q \Sigma) \geq 0$, and $Z_0 \succeq L \Sigma^{-1} L^\top \succeq 0$ with $R \succ 0$ gives $\text{tr}(R Z_0) \geq 0$; hence the objective is ≥ 0 . Slater (Lemma 2) for a bounded-below convex conic program implies zero duality gap and dual attainment by the conic strong-duality theorem [3, Thm. 2.4.1], [4, §5.9]. Primal attainment is exhibited constructively in Proposition 1. $\square$

4 Uniqueness of the primal optimizer via the Riccati solution

We now identify the primal optimum of (2) with the LQR solution and prove (A2). Throughout, “the control DARE” is

$$P = A_\theta^\top P A_\theta - A_\theta^\top P B_\theta (R + B_\theta^\top P B_\theta)^{-1} B_\theta^\top P A_\theta + Q, \quad (7)$$

with the associated gain and closed loop

$$K^* = -(R + B_\theta^\top P B_\theta)^{-1} B_\theta^\top P A_\theta, \quad A_{\text{cl}}^* = A_\theta + B_\theta K^*. \quad (8)$$

Lemma 3 (Classical LQR facts). *Under (S2)–(S3), (7) has a unique symmetric solution $P \succeq 0$ for which A_{cl}^* is Schur; the optimal gain K^* in (8) is the unique minimizer of the stationary LQR cost $J(K) = \text{tr}(Q \Sigma_K) + \text{tr}(R K \Sigma_K K^\top)$ over stabilizing K , where Σ_K solves $\Sigma_K = (A_\theta + B_\theta K) \Sigma_K (A_\theta + B_\theta K)^\top + W$; and the minimum equals $\text{tr}(P W)$.*

Proof. Existence and uniqueness of the stabilizing $P \succeq 0$ under (A_θ, B_θ) stabilizable and $(A_\theta, Q^{1/2})$ detectable is the standard DARE theorem, e.g. [8, Ch. 14–15], [6, Ch. 6]. Uniqueness of the optimal gain and the identity $\min_K J(K) = \text{tr}(PW)$ for the process-noise (H_2) formulation are classical, [6, Ch. 5], [7, Ch. 3]; the latter is exactly Eq. (15) of the paper, $J_c = \text{tr}(PW)$. $\square$

Proposition 1 (Primal optimum and uniqueness). *Under Assumption 1, the unique minimizer of (2) is*

$$\Sigma^* = \Sigma_{K^*} \succ 0, \quad L^* = K^* \Sigma^*, \quad Z_0^* = K^* \Sigma^* (K^*)^\top, \quad (9)$$

with K^*, A_{cl}^* from (8). The optimal value is $J_{\text{cont}}(\theta) = \text{tr}(PW)$. Thus Assumption 1(A2) holds and $\Sigma^* \succ 0$.

Proof. Feasibility and value of (9). Because A_{cl}^* is Schur (Lemma 3) and $W \succ 0$, the Lyapunov equation $\Sigma^* = A_{\text{cl}}^* \Sigma^* (A_{\text{cl}}^*)^\top + W$ has the unique solution $\Sigma^* = \sum_{k \geq 0} (A_{\text{cl}}^*)^k W (A_{\text{cl}}^*)^{\top k} \succ 0$. With $L^* = K^* \Sigma^*$ we get $A_\theta \Sigma^* + B_\theta L^* = A_{\text{cl}}^* \Sigma^*$, and the Schur complement of $\mathcal{G}_c(\Sigma^*, L^*; \theta)$ with respect to $\Sigma^* \succ 0$ is $(\Sigma^* - W) - A_{\text{cl}}^* \Sigma^* (A_{\text{cl}}^*)^\top = 0 \succeq 0$; so $\mathcal{G}_c \succeq 0$. With $Z_0^* = K^* \Sigma^* (K^*)^\top = L^* (\Sigma^*)^{-1} (L^*)^\top$ the middle block has Schur complement $0 \succeq 0$. Hence (9) is feasible, and its objective is $\text{tr}(Q \Sigma^*) + \text{tr}(R K^* \Sigma^* (K^*)^\top) = J(K^*) = \text{tr}(PW)$.

Optimality: no feasible point does better. Let (Σ, L, Z_0) be any feasible triple and set $K := L \Sigma^{-1}$ (so $L = K \Sigma$; $\Sigma \succ 0$). The middle block gives $Z_0 \succeq L \Sigma^{-1} L^\top = K \Sigma K^\top$, whence, $R \succ 0$,

$$\text{tr}(R Z_0) \geq \text{tr}(R K \Sigma K^\top). \quad (10)$$

The Schur complement of $\mathcal{G}_c$ with respect to $\Sigma \succ 0$ reads

$$\Sigma - W - (A_\theta \Sigma + B_\theta L) \Sigma^{-1} (A_\theta \Sigma + B_\theta L)^\top = \Sigma - W - (A_\theta + B_\theta K) \Sigma (A_\theta + B_\theta K)^\top \succeq 0, \quad (11)$$

i.e. $\Sigma \succeq (A_\theta + B_\theta K) \Sigma (A_\theta + B_\theta K)^\top + W$. Since $W \succ 0$ and $\Sigma \succ 0$, this Lyapunov inequality forces $A_\theta + B_\theta K$ to be Schur [5, Ch. 5]; hence K is stabilizing and Σ_K is well defined. Subtracting $\Sigma_K = (A_\theta + B_\theta K) \Sigma_K (A_\theta + B_\theta K)^\top + W$ from (11) gives, with $\Delta := \Sigma - \Sigma_K$,

$$\Delta \succeq (A_\theta + B_\theta K) \Delta (A_\theta + B_\theta K)^\top, \quad (12)$$

and iterating a Schur-stable Stein inequality yields $\Delta \succeq 0$, i.e. $\Sigma \succeq \Sigma_K$. As $Q \succeq 0$, $\text{tr}(Q \Sigma) \geq \text{tr}(Q \Sigma_K)$; combined with (10) and $\Sigma \succeq \Sigma_K$,

$$\text{tr}(Q \Sigma) + \text{tr}(R Z_0) \geq \text{tr}(Q \Sigma_K) + \text{tr}(R K \Sigma_K K^\top) = J(K) \geq \text{tr}(PW),$$

the last step by Lemma 3. Hence $\text{tr}(PW)$ is the optimal value, attained by (9).

Uniqueness. Let (Σ, L, Z_0) be optimal, $K = L \Sigma^{-1}$. Chasing the equalities in the display above: $J(K) = \text{tr}(PW)$ forces $K = K^*$ (unique LQR minimizer, Lemma 3). Then A_{cl}^* is Schur and, from $\text{tr}(R Z_0) = \text{tr}(R K^* \Sigma (K^*)^\top)$ with the constraint $Z_0 \succeq K^* \Sigma (K^*)^\top$ and $R \succ 0$, the linear functional $\text{tr}(R \cdot)$ is minimized on $\{Z_0 \succeq K^* \Sigma (K^*)^\top\}$ only at the boundary point, so $Z_0 = K^* \Sigma (K^*)^\top$ is pinned by Σ . It remains to show $\Sigma = \Sigma^*$. With $K = K^*$, $\Delta = \Sigma - \Sigma^* \succeq 0$ satisfies (12), i.e. $\Delta \succeq A_{\text{cl}}^* \Delta (A_{\text{cl}}^*)^\top$, and optimality forces $\text{tr}(Q \Delta) = 0$, hence $Q^{1/2} \Delta = 0$ and $K^* \Delta = 0$ (the latter because $\text{tr}(R K^* \Delta (K^*)^\top) = \text{tr}(R Z_0) - \text{tr}(R K^* \Sigma^* (K^*)^\top) = \text{tr}(PW) - \text{tr}(PW) = 0$ with $R \succ 0$). Using $K^* \Delta = 0$,

$$A_{\text{cl}}^* \Delta = (A_\theta + B_\theta K^*) \Delta = A_\theta \Delta, \quad \text{so} \quad \Delta \succeq A_\theta \Delta A_\theta^\top.$$

Together with $Q^{1/2} \Delta = 0$ and $\Delta \succeq 0$, detectability of $(A_\theta, Q^{1/2})$ forces $\Delta = 0$: the range of Δ is A_θ -invariant (if $\Delta x = 0$ then $0 = x^\top \Delta x \geq \|\Delta^{1/2} A_\theta^\top x\|^2$, so $\Delta A_\theta^\top x = 0$, i.e. $\ker \Delta$ is $A_\theta^\top$ -invariant and range Δ is A_θ -invariant) and lies in $\ker Q^{1/2}$; a nonzero such subspace would be an A_θ -invariant

unobservable subspace of $(A_\theta, Q^{1/2})$, and on it the inequality $\Delta \succeq A_\theta \Delta A_\theta^\top$ makes A_θ nonexpansive, contradicting that detectability confines the unobservable subspace to the Schur-stable part unless it is trivial.¹ Hence $\Delta = 0$, $\Sigma = \Sigma^*$, and the optimizer is unique. $\square$

Proposition 1 establishes (A2) and, with Corollary 1, completes (A1). It also supplies the primal attainment claimed there.

5 Value differentiability and the dual-invariant gradient

We now prove part (iii) of Theorem 1: $V = J_{\text{cont}}$ is C^1 on Θ_0 , and the envelope formula (5) returns the true gradient for *any* optimal dual — so Corollary 1 holds even without (A3).

The engine is that the optimal value is a smooth function of the model data through the Riccati map.

Lemma 4 (Smoothness of the stabilizing DARE solution). *Let $\mathcal{O} \subseteq \mathbb{R}^{r_\theta}$ be the open set on which (A_θ, B_θ) is stabilizable and $(A_\theta, Q^{1/2})$ detectable. On $\mathcal{O}$ the stabilizing solution $P(\theta)$ of (7) is a C^1 (indeed real-analytic) function of θ , and so are $K^*(\theta)$, $A_{\text{cl}}^*(\theta)$, and $\Sigma^*(\theta)$.*

Proof. Write the DARE residual as $\mathcal{R}(P, \theta) = 0$ with $\mathcal{R} : \mathbb{S}^n \times \mathcal{O} \rightarrow \mathbb{S}^n$ given by the right minus left side of (7); $\mathcal{R}$ is C^∞ (rational, with the invertible $R + B_\theta^\top P B_\theta \succ 0$) and, by (S1), C^1 in θ . At a stabilizing root $P = P(\theta)$ the partial Fréchet derivative $\partial_P \mathcal{R}(P, \theta)$ is the Stein (discrete Lyapunov) operator

$$\partial_P \mathcal{R}(P, \theta)[X] = X - (A_{\text{cl}}^*)^\top X A_{\text{cl}}^*,$$

which is invertible precisely because A_{cl}^* is Schur (its eigenvalues are $1 - \lambda_i \bar{\lambda}_j$ with $|\lambda_i| < 1$, all nonzero). The implicit function theorem yields a unique C^1 branch $\theta \mapsto P(\theta)$ through the stabilizing root; analyticity follows from the analytic IFT. Then K^*, A_{cl}^* are C^1 by (8), and $\Sigma^*(\theta)$ solves a Lyapunov equation with C^1 Schur-stable data, hence is C^1 . $\square$

Proposition 2 (C^1 value). *Under Assumption 1, $J_{\text{cont}}(\theta) = \text{tr}(P(\theta)W(\theta))$ is C^1 on Θ_0 , with*

$$\nabla_\theta J_{\text{cont}} = \langle P, \partial_\theta W \rangle + \langle \Sigma^*, \partial_\theta Q \rangle + \langle K^* \Sigma^* (K^*)^\top, \partial_\theta R \rangle + 2 \langle P A_{\text{cl}}^* \Sigma^*, \partial_\theta A_\theta \rangle + 2 \langle P A_{\text{cl}}^* \Sigma^* (K^*)^\top, \partial_\theta B_\theta \rangle, \quad (13)$$

i.e. the closed-form gradients $\partial J_c / \partial(A, B, W, Q, R)$ of Eq. (18) of the paper.

Proof. C^1 -ness is immediate from Proposition 1 ($J_{\text{cont}} = \text{tr}(PW)$) and Lemma 4 together with (S1). The explicit partials (13) are obtained by differentiating $\text{tr}(PW)$ using the DARE and the Lyapunov identity $\Sigma^* = A_{\text{cl}}^* \Sigma^* (A_{\text{cl}}^*)^\top + W$; the terms coincide with the Riccati/Lyapunov formulas quoted in Section 5 of the paper. (We do not use (13) below; it is recorded to confirm the envelope and Riccati routes agree.) $\square$

The next result is the crux. It shows the envelope right-hand side is the same for every optimal dual, so “read the gradient off the returned dual” is valid irrespective of dual non-uniqueness.

Theorem 2 (Dual-invariant envelope gradient). *Under Assumption 1, let $\mathcal{D}^*(\theta)$ be the (nonempty, compact, convex) set of optimal duals of (2). Then for every $(\mu^*, S^*) \in \mathcal{D}^*(\theta)$,*

$$\nabla_\theta L(z^*, \mu^*, S^*, \theta) = \nabla_\theta J_{\text{cont}}(\theta), \quad (14)$$

¹Concretely, when $Q \succ 0$ the step is immediate: $\text{tr}(Q\Delta) = 0$ with $Q \succ 0$, $\Delta \succeq 0$ gives $\Delta = 0$. The interior-optimum studies of the paper use $Q \succ 0$, so this simplest case already covers them; the detectability argument extends it to $Q \succeq 0$.

where $z^* = (\Sigma^*, L^*, Z_0^*)$ is the unique primal optimizer. In particular the envelope formula (5), and hence the LMI-dual gradients of Corollary 1 (Eqs. (12)–(13) of the paper), evaluated at any solver-returned optimal dual, equal the true design gradient.

Proof. By Slater (Lemma 2) the dual optimal set $\mathcal{D}^*(\theta)$ is nonempty, convex and compact, and by (A2) the primal minimizer z^* is unique. Under these conditions the value function of the convex program (4) is directionally differentiable in θ with

$$J'_{\text{cont}}(\theta; d) = \min_{(\mu, S) \in \mathcal{D}^*(\theta)} \langle \nabla_{\theta} L(z^*, \mu, S, \theta), d \rangle, \quad d \in \mathbb{R}^{r_{\theta}}, \quad (15)$$

by the perturbation-analysis / Danskin-type theorem for optimal values of convex problems with a unique primal solution [3, Thm. 4.24, Prop. 4.22] (this is exactly the directional formula quoted in Remark 1 of the paper).

By Proposition 2, J_{cont} is differentiable at θ , so its directional derivative is *linear* in d : $J'_{\text{cont}}(\theta; d) = \langle \nabla_{\theta} J_{\text{cont}}, d \rangle$ for all d . Combining with (15),

$$\min_{(\mu, S) \in \mathcal{D}^*} \langle \nabla_{\theta} L(z^*, \mu, S, \theta), d \rangle = \langle \nabla_{\theta} J_{\text{cont}}, d \rangle \quad \text{for all } d. \quad (16)$$

Fix any $(\mu^*, S^*) \in \mathcal{D}^*$ and abbreviate $a := \nabla_{\theta} L(z^*, \mu^*, S^*, \theta)$ and $g := \nabla_{\theta} J_{\text{cont}}$. Since (μ^*, S^*) is a feasible point of the min in (16), $\langle a, d \rangle \geq \min_{\mathcal{D}^*} \langle \cdot, d \rangle = \langle g, d \rangle$ for all $d \in \mathbb{R}^{r_{\theta}}$. A linear functional that dominates another for every d (including $-d$) must equal it; therefore $a = g$. As (μ^*, S^*) was arbitrary in $\mathcal{D}^*$, (14) holds for every optimal dual. Finally $\nabla_{\theta} L(z^*, \mu^*, S^*, \theta)$ is precisely the right-hand side of the envelope formula (5), whose block-partitioned specialization is Eq. (12) of the paper ($\partial J_{\text{cont}} / \partial A_{\theta} = -2S_{12}\Sigma^*$, $\partial J_{\text{cont}} / \partial B_{\theta} = -2S_{12}(L^*)^{\top}$, and the objective terms of Eq. (16)); these therefore return the exact gradient. $\square$

Remark 1 (Why this suffices for the algorithm). The outer method of the paper needs $\nabla_{\theta} J_{\text{stoch}}$ only to take a descent step. Theorem 2 guarantees that whatever optimal dual the interior-point solver returns, contracting it against the model Jacobians $\partial_{\theta} D$ (Eq. (13)) yields the true gradient. Thus Corollary 1 is *operationally* valid under (S1)–(S3) alone. Strict complementarity (A3) is needed only if one wants the stronger structural statement that the dual is literally unique; we turn to it next.

6 Strict complementarity, nondegeneracy, and dual uniqueness (A3)

Assumption 1(A3) asks for a nondegenerate, strictly complementary primal–dual solution, which is the standard sufficient condition for the dual to be a singleton and for the solution map $\theta \mapsto (z^*, \mu^*, S^*)$ to be differentiable [1, 2]. We record what it means here and when it holds.

Write the active Lyapunov LMI at the optimum, $\mathcal{G}^* := \mathcal{G}_c(\Sigma^*, L^*; \theta) \in \mathbb{S}^{2n}$, with dual block $S^* = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^{\top} & S_{22} \end{bmatrix} \succeq 0$. By Proposition 1 the Schur complement of $\mathcal{G}^*$ vanishes and its $(2, 2)$ block $\Sigma^* \succ 0$, so

$$\text{rank } \mathcal{G}^* = n, \quad \ker \mathcal{G}^* = \text{range} \begin{bmatrix} I_n \\ -(A_{\text{cl}}^*)^{\top} \end{bmatrix}, \quad (17)$$

the second identity because $\mathcal{G}^* \begin{bmatrix} I \\ -(A_{\text{cl}}^*)^{\top} \end{bmatrix} = \begin{bmatrix} \Sigma^* - W - A_{\text{cl}}^* \Sigma^* (A_{\text{cl}}^*)^{\top} \\ A_{\text{cl}}^* \Sigma^* - \Sigma^* (A_{\text{cl}}^*)^{\top} (\dots) \end{bmatrix} = 0$ at the tight solution.

Proposition 3 (Strict complementarity $\Rightarrow$ unique dual). *Suppose that at the optimum of (2) the primal–dual pair is strictly complementary, i.e. the multiplier S^* of the active Lyapunov LMI satisfies*

$$S^* \mathcal{G}^* = 0, \quad \text{rank } S^* + \text{rank } \mathcal{G}^* = 2n \quad (\text{so } \text{rank } S^* = n), \quad (18)$$

together with the analogous conditions on the middle-block and $\Sigma \succ 0$ constraints, and that the primal solution is nondegenerate in the sense of [1]. Then the dual (μ^, S^*) is unique, so Assumption 1(A3) holds; combined with Sections 2–5, Assumption 1(A1)–(A4) holds in full and Theorem 1 applies verbatim.*

Proof. For a semidefinite program, primal nondegeneracy at z^* implies the dual optimizer is unique, and strict complementarity implies the primal optimizer is unique and the primal–dual map is differentiable; these are the AHO/Shapiro characterizations [1, Thm. 10, Thm. 15], [2, Thm. 4.1]. The uniqueness of z^* we already have (Proposition 1); the added hypothesis supplies uniqueness of (μ^*, S^*) . Hence (A3), and with (A1),(A2),(A4) proved earlier, all of Assumption 1 holds, so Theorem 1 applies and its conclusion (5) is the (now unique) envelope gradient. $\square$

Remark 2 (Verifiability and the a-posteriori check). Condition (18) is checkable at the returned solution: form $\mathcal{G}^*$ and S^* , verify $\mathcal{G}^* S^* = 0$ and $\text{rank } \mathcal{G}^* + \text{rank } S^* = 2n$ numerically. Because $\text{rank } \mathcal{G}^* = n$ *always* holds at the optimum (17), strict complementarity reduces to the single condition $\text{rank } S^* = n$ with $S^* \mathcal{G}^* = 0$ — exactly the nondegeneracy that the paper verifies a posteriori through the finite-difference gradient check (Remark 1). In every numerical study of the paper this held, so (A3), and thus Assumption 1 in full, was met; where it might fail, Remark 1 guarantees the returned-dual gradient is still exact and the outer method degrades at worst to a subgradient step.

7 The estimation SDP by duality

The estimation SDP (3) is the transpose dual of the control SDP: it is obtained from (2) under the substitution

$$A_\theta \mapsto A_\theta^\top, \quad B_\theta \mapsto C_\theta^\top, \quad W \mapsto Q, \quad Q \mapsto W, \quad R \mapsto V, \quad (\Sigma, L, Z_0) \mapsto (P^\varepsilon, -F, Z_1). \quad (19)$$

Under (19) the objective $\text{tr}(Q\Sigma) + \text{tr}(RZ_0)$ becomes $\text{tr}(WP^\varepsilon) + \text{tr}(VZ_1)$, and the constraint blocks $\mathcal{G}_c$ and $\begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix}$ become $\mathcal{G}_e$ and $\begin{bmatrix} Z_1 & F^\top \\ F & P^\varepsilon \end{bmatrix}$, respectively. The system-theoretic hypotheses map correctly:

$$(A_\theta, B_\theta) \text{ stabilizable} \leftrightarrow (A_\theta^\top, C_\theta^\top) \text{ stabilizable} \equiv (A_\theta, C_\theta) \text{ detectable},$$

and $(A_\theta, Q^{1/2})$ detectable $\leftrightarrow (A_\theta^\top, W^{1/2})$ detectable $\equiv (A_\theta, W^{1/2})$ stabilizable, which is *automatic* because $W \succ 0$ makes $(A_\theta, W^{1/2})$ controllable. The role of “ $W \succ 0$ ” in the control proof (guaranteeing $\Sigma^* \succ 0$ and forcing the closed loop Schur) is played here by “ $Q \mapsto W \succ 0$ ”, and the role of “ $R \succ 0$ ” by “ $V \succ 0$ ”. Hence:

Corollary 2 (Estimation SDP). *Under Assumption 1, all conclusions of Theorem 1(i)–(iv), Lemma 2, Proposition 1, Proposition 2, and Theorem 2 hold for the estimation SDP (3) under the substitution (19). In particular (3) has a Slater point, a unique primal optimizer $P^{\varepsilon*} \succ 0$ equal to the stabilizing solution of the filter DARE*

$$P^\varepsilon = A_\theta^\top P^\varepsilon A_\theta - A_\theta^\top P^\varepsilon C_\theta^\top (V + C_\theta P^\varepsilon C_\theta^\top)^{-1} C_\theta P^\varepsilon A_\theta + W,$$

a C^1 value, and a dual-invariant envelope gradient; consequently the estimation-side design gradients of Corollary 1 (the parallel of Eqs. (12)–(13) in C_θ, W, V , and Eq. (17)) read from the returned LMI duals are exact.

Proof. Every step of Sections 2–6 is invariant under the linear relabeling (19); the hypotheses required there — stabilizability of the “ (A, B) ” pair, detectability of the “ $(A, Q^{1/2})$ ” pair, and positive definiteness of the “ W ” and “ R ” data — are exactly the detectability of (A_θ, C_θ) , the (automatic) stabilizability of $(A_\theta, W^{1/2})$, and $Q \mapsto W \succ 0$, $R \mapsto V \succ 0$ supplied by Assumption 1. The claims transfer verbatim. $\square$

8 Discussion

Theorem 1 converts Corollary 1 into a statement under textbook hypotheses. The logical structure is worth isolating:

- **Slater (A1) $\Leftarrow$ stabilizability/detectability + $W, V \succ 0$.** The Slater point is not generic: it is built from a stabilizing feedback (control side) or output injection (estimation side), i.e. from exactly the controllability/observability data. Positive-definite process/measurement noise provides the strict margin.
- **Uniqueness (A2) $\Leftarrow$ the DARE.** The covariance-form SDP is an exact convex lift of the LQR/Kalman problem; its unique optimizer is the Riccati solution. Detectability of $(A_\theta, Q^{1/2})$ (automatic if $Q \succ 0$) and (A_θ, C_θ) pins the optimizer.
- **Gradient correctness needs *less than* (A3).** Because the value $\text{tr}(PW)$ is a smooth function of the Riccati solution (Lemma 4), the value is C^1 regardless of dual non-uniqueness, and the envelope right-hand side is dual-invariant (Theorem 2). This is why the finite-difference check in the paper always matched the dual-read gradient, and why the method is robust even at solutions where strict complementarity is borderline.
- **(A3) is the exact upgrade to dual uniqueness.** Strict complementarity, which reduces here to $\text{rank } S^* = n$ at the (always rank- n) active Lyapunov LMI, is necessary and sufficient in the AHO sense for the dual to be a singleton, recovering Assumption 1 as written.

In short: *controllability and observability (stabilizability/detectability), with positive-definite noise and control weights, imply Assumption 1(A1)–(A2) and the differentiability and exactness of the design gradient; strict complementarity, verifiable a posteriori, supplies the remaining (A3).* Theorem 1 therefore applies to the LQG co-design SDPs under standard system-theoretic hypotheses, and Corollary 1 holds unconditionally under (S1)–(S3).

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